Mrugesh Thesiya

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Full-stack engineer who ships products real people depend on. Took PDEU's management portal from a first requirements conversation to production, and it now serves 450+ students and 40+ faculty. Owns the whole loop: talking to users, building, deploying, and supporting. Also building Vajra, an open-source security tool for AI coding agents, and works daily with those agents to ship faster.

Projects

- **PDEU Management Portal (portal.pdpu.ac.in)** — In production
  TypeScript, React, Node.js, Express, Appwrite (self-hosted), LDAP, Docker — *May 2024 – Present*
  - Own the portal end to end, from requirements to deployment and support: one product for industrial training and for minor, major, and comprehensive projects, used by **450+ students and 40+ faculty**.
  - Replaced manual submission and tracking with one workflow: team and mentor allocation, submissions with deadlines, evaluation and grading, training applications, document uploads, and status reports.
  - Ran requirements gathering, demos, and feedback rounds directly with faculty, administrators, and students; triage their bug reports and feature requests and ship the fixes.
  - Integrated sign-in with the campus LDAP directory by working with the university IT department, and built role-based access for SuperAdmin, faculty, and mentors.
  - Rebuilt the original Next.js app on React and Node.js in a pnpm and Turborepo monorepo, with self-hosted Appwrite for auth, database, and storage; containerized with Docker and deployed on the university's own server.
- **Vajra – Sandbox for AI Coding Agents** — Open source, Apache 2.0
  Rust, TypeScript, napi-rs, pnpm, Turborepo
  - Building an open-source sandbox that confines AI CLI agents (Claude Code, Codex, OpenCode) to a single project, so they can work on code without reaching secrets elsewhere on the machine.
  - Wrote the native core in Rust, exposed to Node through napi-rs, enforcing filesystem confinement with Landlock on Linux and Seatbelt on macOS.
  - Hides `.env` files from file listings and automatically redacts secret values in agent output.
  - Shipped a CLI for multi-agent task execution with developer, master, and worker roles and per-file permissions, across five packages in one monorepo.
- **Open Elective Choice Filling System** — PDEU
  Go, React, REST APIs, Docker
  - Designed and built a Go backend that collects open elective choices from every student of a PDEU school in one pass, run department by department, with a React front end for students and coordinators.
  - Built the REST API in Go with request validation, so each student can submit only electives offered to their department and cannot submit duplicate or conflicting choices.
  - Handled many students submitting at the same time with Go's concurrency primitives and safe writes, so choices are never lost or overwritten during a filling window.
  - Built coordinator controls to configure electives, seat limits, and filling windows per department, and to export the collected choices for allocation.
  - Secured the API with authenticated access and role checks, and containerized the service with Docker for deployment.

Experience

- **Web Development Committee, PDEU** — Gandhinagar, India
  Committee Lead — *July 2024 – Present*
  - Lead a team of student developers building production software for the university under faculty mentorship.
  - Teach a weekly engineering session to **60+ students** on AI coding agents and context engineering, learning with AI, and backend architecture, authentication, databases, and infrastructure.
  - Write and publish the session guides at wdc-guides.pages.dev.
- **Pandit Deendayal Energy University (PDEU)** — Hybrid
  Tech Intern — *May 2024 – July 2024*
  - Scoped and built the first version of the project submission portal with faculty stakeholders in Next.js and Appwrite; it grew into the production portal above.

Skills

- **Languages**: TypeScript, JavaScript, Go, Python, Java
- **Web**: React.js, Next.js, Node.js, Express.js, Astro, Tailwind CSS, REST APIs
- **Data and Auth**: Appwrite (self-hosted), MySQL, LDAP, role-based access control
- **Infrastructure**: Docker, Nginx, PM2, Git, Linux servers, AWS, DigitalOcean
- **AI-Assisted Development**: Claude Code and other coding agents, context engineering, agent sandboxing

Education

- **Pandit Deendayal Energy University (PDEU)** — Gandhinagar, India
  B.Tech. Computer Science and Engineering; CGPA: 9.44 — *July 2023 – Present*